Hierarchical Shell Manifold Theory: Minimal Operator Core, Modular Extensions, and the Predictivity Gap

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Abstract

Hierarchical Shell Manifold Theory (HSMT) began as an attempt to organize effective physics through a hierarchy of dynamically coupled geometric sectors. Subsequent development has substantially narrowed that proposal. Several structures that initially appeared potentially fundamental—including an absolute grade assignment, a preferred Gaussian width, specific dimensional identifications, vacuum-selection mechanisms, and possible matter or cosmological interpretations—have either been demoted to conditional realizations or closed by explicit nonselection and no-go results.

This paper presents the post-maturation form of HSMT without introducing new physics. We distinguish the public geometric architecture from a theorem-minimal operator substrate. The latter consists of an affine-discrete graded Hilbert space, a self-adjoint microscopic generator, and an affine clock, and is mathematically equivalent to generic affine-graded quantum mechanics. HSMT-specific identity therefore does not reside in this substrate alone. It resides in a modular architecture combining grade-wise geometric realization, nontrivial intergrade dynamics, and exact projection into effective sectors.

We classify the framework into four maturity levels: generic substrate, architectural HSMT, constitutive HSMT models, and an empirically distinctive physical HSMT theory. Current HSMT reaches the architectural level and contains several conditional constitutive branches, but no established empirical completion. We summarize the principal positive theorems, conditional constructions, no-go results, demoted historical claims, and the optional P-CYCLE extension.

The central predictivity gap is now precise: HSMT presently lacks a physically selected and operationally identifiable grading together with a non-generic measurable relation among its selected sectors. An admissibility envelope for any future solution of this problem is stated explicitly.

The resulting framework is smaller than earlier versions of HSMT, but more sharply defined. The present work should therefore be read as a foundational status paper: it records what has survived, what has been shed, what remains conditional, and what would be required for HSMT to become an empirically distinctive physical theory.

1 Introduction

Hierarchical Shell Manifold Theory was motivated by the possibility that familiar effective physics could arise from a hierarchy of geometric sectors whose interactions are more fundamental than the physics observed within any one sector. The September 2026 reference framework records

the public geometric and operator architecture on which the present paper builds [1]. Subsequent inverse-spectral and no-go work sharpened the identifiability boundary of that architecture [2].

Early formulations explored several possible identifications at once: hierarchy grade with effective dimension, special Gaussian coupling widths, spectral scaling chains, emergent temporal structure, matter-like algebraic sectors, geometric vacuum selection, and possible cosmological consequences. The exploratory phase generated useful mathematical structures, but it also made a stricter logical classification necessary. A framework with many compatible constructions can appear more predictive than it is unless one separates

(i) what follows by theorem,

(ii) what follows only after an additional constitutive assumption,

(iii) what has been excluded by a no-go or nonselection result,

(iv) what remains historical motivation, and

(v) what is operationally connected to measurement.

The post-maturation programme enforced these distinctions. The result is not a larger theory. It is a smaller and more modular one.

Several mechanisms once considered possible physical selectors failed to select anything. Several mathematically coherent structures remain without a physical bridge. Several negative results now place stronger constraints on future theory construction than the positive constructions that originally motivated them.

The purpose of this paper is therefore not to add a new mechanism. It is to answer a foundational question:

What is HSMT after every result is assigned its correct logical status?

The answer requires a clean separation between

1. a generic graded quantum substrate,
2. the HSMT architectural package, and
3. conditional or optional modules built on that architecture.

After this separation, the main unresolved problem becomes precise: HSMT currently lacks a physical principle that selects the grading operationally and simultaneously imposes at least one non-generic measurable intergrade relation.

2 The Frozen Public Architecture

The public HSMT architecture remains

$$\mathcal{M} = \bigsqcup_N \mathcal{M}_N \longrightarrow D_\rho, D_{\mathrm{sp}} \longrightarrow \text{states/packets} \longrightarrow \langle N|M\rangle \longrightarrow \mathcal{D} \longrightarrow P_N \longrightarrow H_{\mathrm{eff}}. \tag{1}$$

This architecture remains useful because it displays the intended geometric interpretation inherited from the September reference framework [1].

The first object is the graded arena

$$\mathcal{M} = \bigsqcup_N \mathcal{M}_N, \tag{2}$$

where N labels a hierarchy sector. The realization-dependent operators D_ρ and D_{sp} encode radial and spatial structure. States and packets occupy the corresponding state space, and overlaps encode relations among selected states. A reference operator $\mathcal{D}$ supplies the associative operator laboratory. Projection by P_N then defines an effective sector described by H_{eff}.

The measurement layer is downstream:

$$Q_N(\ell) \longrightarrow d_N(\ell) = \frac{d \ln Q_N(\ell)}{d \ln \ell}. \tag{3}$$

Equation (3) is meaningful only after $Q_N(\ell)$ is operationally specified. Scale-dependent and multifractional dimension constructions provide broad mathematical context for such running quantities, but do not by themselves identify an HSMT grade with a physical dimension [13].

Three interpretive firewalls remain fixed:

$$N \neq d, \tag{4}$$

$$\text{shell} \neq \text{dimension}, \tag{5}$$

and no equality is assumed among

$$d_{\mathrm{geom}}, \qquad d_{\mathrm{res}}, \qquad d_{\mathrm{spec}}, \qquad d_{\mathrm{dyn}}. \tag{6}$$

The hierarchy label therefore cannot silently be promoted to a physical dimension.

3 The Theorem-Minimal Operator Substrate

The public architecture contains more named structure than is required for the exact operator results.

Let Γ be a discrete affine grade set. It possesses relative separation but no preferred origin. An integer coordinate $N : \Gamma \to \mathbb{Z}$ therefore admits the relabeling

$$N \mapsto N + c. \tag{7}$$

The theorem-minimal state space is

$$\mathcal{H} = \bigoplus_{\gamma \in \Gamma} \mathcal{H}_\gamma. \tag{8}$$

Once this decomposition is declared, the orthogonal projectors P_γ are derived.

Let

$$H = H^\dagger \tag{9}$$

be a densely defined self-adjoint microscopic generator. Standard self-adjoint perturbation theory supplies the appropriate control when the intergrade hopping is bounded [3]. A nontrivial graded realization may contain intergrade blocks

$$P_\gamma H P_{\gamma'} \neq 0, \qquad \gamma \neq \gamma'. \tag{10}$$

For dynamical interpretation, introduce an affine clock t and

$$U(t) = e^{-itH}. \tag{11}$$

The minimal dynamical substrate is therefore

$$\mathfrak{C}^{\min}_{\mathrm{HSMT}} = \left(\Gamma, \mathcal{H} = \bigoplus_{\gamma \in \Gamma} \mathcal{H}_\gamma, H, t \right). \tag{12}$$

For static spectral statements, t may be omitted.

3.1 Exact projected dynamics

For a selected projector P and $Q = I - P$, the Feshbach–Schur reduction [4, 5] gives

$$P(z - H)^{-1} P = \left[z - PHP - PHQ(z - QHQ)^{-1} QHP \right]^{-1}. \tag{13}$$

The corresponding effective operator is

$$H_{\mathrm{eff}}(z) = PHP + PHQ(z - QHQ)^{-1} QHP. \tag{14}$$

The second term in (14) is the exact self-energy or memory contribution generated by the eliminated sector. This structure is closely related to standard projection-operator approaches to reduced dynamics [6, 7, 8].

States, overlaps, grade projectors, effective operators, and memory terms are therefore derived once the grading and microscopic generator are given.

4 Generic-Core Equivalence

Compression exposes a basic structural fact.

Theorem 4.1 (Generic-core equivalence)**.** *The theorem-minimal HSMT substrate*

$$\left(\Gamma, \bigoplus_{\gamma} \mathcal{H}_\gamma, H, t \right)$$

contains exactly the mathematical data of generic affine-discrete graded quantum mechanics.

Proof. A generic affine-discrete graded quantum system consists of a discrete affine index set, a graded Hilbert space, a self-adjoint generator not required to preserve the grading, and an affine evolution parameter. These are exactly the data in (12). Conversely, every realization of (12) defines such a graded quantum system.

Taking grade-preserving unitary intertwiners, up to allowed affine relabeling, as the natural morphisms produces the corresponding equivalence of model classes. □

Corollary 4.2. *Any theorem proved using only the theorem-minimal substrate is also a theorem of generic affine-graded quantum mechanics.*

This includes Hilbert-space state structure, block decompositions, self-adjoint evolution, resolvent identities, Schur complements, Feshbach reduction, self-energy and memory terms, spectral measures, standard min–max results, and Hellmann–Feynman differentiation.

These are essential consistency tools. They are not uniquely HSMT.

Core interpretation. The theorem-minimal core is the common mathematical substrate on which HSMT modules are built. No novelty claim is attached to the substrate alone.

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5 Where HSMT Identity Begins

Let M_{GEO} denote grade-wise geometric realization and M_{K} a declared intergrade coupling module.

The smallest package preserving the historical HSMT architectural concept is

$$\mathfrak{A}_{\mathrm{HSMT}} = \text{MC-Core} + M_{\mathrm{GEO}} + M_{\mathrm{K}}. \tag{15}$$

Its conceptual chain is

$$\text{graded geometry} \longrightarrow \text{cross-grade dynamics} \longrightarrow \text{projected effective sector}. \tag{16}$$

A geometric hierarchy without intergrade dynamics is only a collection of graded geometries. A coupled graded quantum system without grade-wise geometric realization is only a multilevel graded quantum model. Their combination, together with exact projection, defines the present architectural identity of HSMT.

There is a further distinction. If the intergrade operator is completely arbitrary, then (15) defines an architecture family rather than a constitutive theory. A constitutive HSMT model requires a restricted law governing intergrade structure.

6 Four Levels of Theory Maturity

Level 0: Generic substrate

$$\text{MC-Core}.$$

This is generic affine-graded quantum mechanics.

Level 1: Architectural HSMT

$$\text{MC-Core} + M_{\mathrm{GEO}} + M_{\mathrm{K}}.$$

This is the framework-level HSMT architecture.

Level 2: Constitutive HSMT model

Level 1 plus one or more restricted module laws. Examples include a specified intergrade kernel, locality law, metric law, refinement law, matter branch, gravity branch, or optional extension.

Level 3: Empirically distinctive physical HSMT theory

Level 2 plus

1. an operationally admissible physical grading selector,
2. a physical bridge to preparations, controls, and readouts, and
3. at least one non-generic out-of-sample measurable relation.

Current theory level. HSMT is presently a Level-1 framework with multiple conditional Level-2 branches and no established Level-3 completion.

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7 Modular Organization

The post-maturation theory is most transparent when written schematically as

$$\text{HSMT} = \text{MC-Core} + \text{declared modules}. \tag{17}$$

7.1 M_{GEO}: geometric realization

This module restores the shell-manifold interpretation

$$\mathcal{M} = \bigsqcup_{\gamma \in \Gamma} \mathcal{M}_\gamma \tag{18}$$

and realization-dependent geometric operators.

7.2 M_{K}: intergrade constitutive law

This module restricts the off-diagonal blocks of H. A historical realization is

$$K_{\gamma\gamma'} \propto \exp\left[-\frac{(\gamma - \gamma')^2}{2\sigma^2}\right]. \tag{19}$$

Gaussianity is not a theorem of the minimal core, and no preferred σ is selected there.

7.3 M_{L}: relational locality

This module declares primitive relational locality rather than deriving a unique neighborhood relation from fully symmetric support data.

7.4 M_{ME}: metric–measure–energy

This module adds positive measure, conductance, metric, and Dirichlet-energy data to the relational support. Standard graph and Dirichlet-form theory provide the mathematical background for these structures [10, 11].

7.5 M_{R}: refinement

This module governs controlled changes of relational complexes and their associated state/measure transport.

7.6 M_{GR}: gravity-response branches

This module contains declared geometry-dependent spectral responses. The present positive spectral branch yields a controlled vacuum obstruction, not Einstein gravity.

7.7 M_{MAT}: matter and internal algebra

This module contains conditional Clifford-like, internal-symmetry, multiplicity, scalar, overlap, and seesaw constructions. Noncommutative and spin-geometric structures provide broad mathematical context, but no Standard Model identification is presently derived [9, 12].

7.8 M_{OP}: operational bridge

This module maps internal structures to preparations, controls, readouts, and the affine clock. It adds no ontology by itself, but is necessary for empirical interpretation.

7.9 M_{PC}: P-CYCLE

P-CYCLE is an optional extension and is not part of HSMT-Core.

8 Frozen Positive Results

The results in this section survive as derived statements under their declared hypotheses.

8.1 Self-adjoint operator control

For bounded intergrade hopping, standard Kato–Rellich control provides a self-adjoint microscopic generator [3]. This is mathematically important but belongs to generic operator theory.

8.2 Exact projection

Equation (13) gives exact projected resolvent dynamics. Hidden sectors generate the exact self-energy

$$\Sigma(z) = PHQ(z - QHQ)^{-1}QHP. \tag{20}$$

8.3 Affine grade freedom

No absolute grade origin is selected by the current architecture. The shift

$$N \mapsto N + c \tag{21}$$

therefore remains a label freedom unless an additional physical anchor is supplied.

8.4 Grade and dimension

Nothing in the theorem-minimal architecture implies

$$N = d. \tag{22}$$

This remains a permanent interpretive firewall.

8.5 Support differential structure

Once a relational support complex is specified, its incidence/exterior differential is canonical. Adjoints and Hodge structures require additional metric data.

8.6 Operational stabilizer

Let B_a be operational contacts and C_α calibrated controls. Define

$$\mathcal{G}_{\mathrm{op}} = \left\{ U : UB_a = B_a, \quad UC_\alpha U^\dagger = C_\alpha \right\}. \tag{23}$$

Internal structures related by an element of $\mathcal{G}_{\mathrm{op}}$ cannot be distinguished by those operations.

9 Conditional Historical Kernel Structure

The Gaussian realization (19) remains mathematically coherent.

The historical candidate

$$\sigma_0 = \frac{\sqrt{2}}{4} \tag{24}$$

exhibits nearest-neighbor dominance within the declared realization. This is a numerical property of that realization, not a derivation of a physical constant.

Likewise, the historical implication chain beginning with

$$\Omega(y) \sim y^{5/6} \quad \text{or} \quad y^{7/6} \tag{25}$$

and passing through

$$-\frac{35}{144y^2} \to \nu_B = \frac{1}{12} \to \mathrm{DtN} \sim A^{1/12} \to M^{-1/6} \tag{26}$$

and subsequently

$$d\mu(M) \sim M^{-1/3} dM, \qquad \Sigma \sim p^2 E^{-1/3}, \qquad E \sim p^{3/2} \tag{27}$$

remains conditional on its input scaling assumptions. It is not selected by HSMT-Core.

10 Locality and Metric–Measure Structure

Fully symmetric or exchangeable support data do not by themselves select a preferred neighborhood relation through deterministic equivariant construction. Locality must therefore enter through a declared primitive or another independently motivated relational structure.

Once a support graph or complex is given, the metric–measure–energy module may introduce

$$(\ell_e, \mu_v, c_e) \tag{28}$$

together with a positive Dirichlet form. These structures provide a controlled route to relational geometry. They do not presently identify the physical geometry of Nature.

11 Temporal Structure

The minimal temporal interface consists of an affine parameter t and

$$U(t) = e^{-itH}. \tag{29}$$

Grade is not time.

A causal interpretation can be constructed conditionally after selection of a suitable principal symbol. That symbol is not selected by the minimal core.

A separate negative result is important: a closed finite recurrent unitary system does not supply a globally monotone intrinsic Lyapunov arrow solely from reversible microscopic dynamics. Temporal ordering and thermodynamic arrow must therefore remain distinct.

12 Gravity Programme and Vacuum Obstruction

One investigated geometry-dependent branch has

$$A(q) = m_f^2 I + B^T W(q) B. \tag{30}$$

Under

$$W(q) = e^q W_0, \tag{31}$$

the zero-point energy is

$$E_0(q) = \frac{1}{2} \operatorname{Tr} \sqrt{A(q)}. \tag{32}$$

For the declared positive branch,

$$\frac{\partial E_0}{\partial q_e} = \frac{1}{4} w_e \, b_e^T A^{-1/2} b_e > 0. \tag{33}$$

There is therefore no finite unconstrained stationary point in this branch. This is an internal no-go result, not a derivation of physical gravity.

Subsequent positive q-independent refinement mixing does not reverse the monotonicity. Canonical positive Gaussian overlap structures likewise do not supply the missing stabilizing sign. These results constrain future vacuum proposals.

13 The Optional P-CYCLE Extension

P-CYCLE was introduced explicitly as a new primitive extension rather than a core derivation.

For

$$H_1(\mathfrak{C}; \mathbb{R}) = Z_1 / B_1, \tag{34}$$

define a cycle metric G and phase space

$$T^* H_1. \tag{35}$$

The frozen oscillator Hamiltonian is

$$H_{\mathrm{cyc}} = \frac{\omega_c}{2}\left[\pi^T G^{-1}\pi + h^T G h\right]. \tag{36}$$

After metric standardization, the spectrum is

$$E_{\mathbf{n}}^{\mathrm{cyc}} = \omega_c\left(|\mathbf{n}| + \frac{b_1}{2}\right). \tag{37}$$

The spectrum is therefore independent of G.
Under

$$G(q) = e^{-q}G_0, \tag{38}$$

the Hamiltonian is unitarily equivalent to the fixed isotropic oscillator. Hence

$$\frac{\partial E_{\mathbf{n}}^{\mathrm{cyc}}}{\partial q} = 0. \tag{39}$$

P-CYCLE therefore does not cure the tested GC vacuum runaway. That vacuum role is closed.
P-CYCLE remains a coherent optional structural extension. It is not identified with electromagnetism, gauge structure, dark matter, dark energy, or any known field.

14 Operational Analysis

The empirical programme begins with the operational tuple

$$\mathfrak{O} = (\mathcal{P}, \mathcal{C}, \mathcal{R}, t), \tag{40}$$

representing preparations, controls, readout algebra, and affine clock.
For a contact map B,

$$M_B(z) = B^\dagger (z - H)^{-1} B. \tag{41}$$

For self-adjoint H and $\operatorname{Im} z > 0$,

$$-\operatorname{Im} M_B(z) = (\operatorname{Im} z)\, B^\dagger (z - H)^{-1}(\bar{z} - H)^{-1} B \geq 0. \tag{42}$$

At large z,

$$M_B(z) = \frac{B^\dagger B}{z} + \frac{B^\dagger H B}{z^2} + O(z^{-3}). \tag{43}$$

These are exact operator consistency tests. They are not uniquely HSMT.

14.1 Contact-stabilizer obstruction

If

$$UB = B \tag{44}$$

and

$$H' = UHU^\dagger, \tag{45}$$

then

$$B^\dagger(z - H')^{-1}B = B^\dagger(z - H)^{-1}B. \tag{46}$$

Contact spectroscopy therefore cannot identify a physical HSMT grading unless the operational algebra removes the stabilizer ambiguity.

15 Controlled Analog Results

A minimal benchmark uses

$$P_3 \longrightarrow C_3 \tag{47}$$

to change the first Betti number from zero to one.

An edge-sector analog Hamiltonian is

$$H_{\text{edge}} = \frac{1}{2}p^T p + \frac{1}{2}x^T \left(\omega_c^2 I + BKB^T\right) x, \qquad K > 0. \tag{48}$$

For

$$h \in \ker B^T, \tag{49}$$

one has

$$BKB^T h = 0, \tag{50}$$

and therefore

$$\Omega_{\text{cyc}} = \omega_c. \tag{51}$$

The multiplicity is

$$\dim \ker B^T = b_1. \tag{52}$$

Positive changes of K preserve this cycle mode exactly.

The analog construction is nontrivial at the realization level. However, a sufficiently general ordinary oscillator network can reproduce the same quadratic dynamics and transfer functions. Controlled analog realization demonstrates mathematical realizability, not fundamental ontology.

16 Natural-Bridge Boundary

A natural physical bridge must be stronger than mathematical resemblance.

A qualifying candidate must possess independently established

1. physical degrees of freedom,
2. measurable topology,
3. contact and readout laws,
4. canonical structure,

5. constitutive dynamics,

6. a bridge not fitted from the proposed test,

7. at least one unused prediction,

8. a conventional comparator, and

9. branch-level falsifiability.

A loop, conserved current, kernel mode, or protected resonance is insufficient by itself.

Broad conceptual candidate classes include conservative oscillator networks, flow networks, mechanical kernel systems, quantum loop/flux systems, and topologically protected wave or material systems. None presently supplies the complete frozen P-CYCLE bridge. The Nature-facing P-CYCLE route is therefore blocked, not falsified.

17 Core Empirical Uniqueness

The most severe empirical audit asked whether any present HSMT-Core statement is simultaneously

1. HSMT-specific, and

2. operationally identifiable.

Generic operator theorems fail the first condition. The physical grading, shell projector, and intergrade kernel fail the second. Running or fractional dimension is not uniquely HSMT. Relational graph locality is not uniquely HSMT. Conditional matter structures do not presently identify observed matter. The gravity branch currently supplies an obstruction rather than a successful Nature-facing law.

Empirical uniqueness result. No currently frozen HSMT-Core relation is a uniquely identified Nature-facing distinguishing prediction.

This is not an experimental falsification of HSMT. It is a classification of its present theory status.

18 The Operational Grading-Selector Envelope

The missing physical principle can be constrained without inventing it.

Let

$$\mathscr{P} = \{P_\gamma\}_{\gamma\in\Gamma} \tag{53}$$

be a complete orthogonal grading,

$$P_\gamma P_{\gamma'} = \delta_{\gamma\gamma'} P_\gamma, \qquad \sum_\gamma P_\gamma = I. \tag{54}$$

Because the grade origin is affine, the physical grading should initially be treated as an equivalence class

$$[\mathscr{P}] \in \mathrm{Grad}_\Gamma(\mathcal{H})/\mathcal{A}_\Gamma. \tag{55}$$

A future selector is schematically

$$\mathscr{S} : \mathcal{X}_{\mathrm{id}} \longrightarrow \mathrm{Grad}_\Gamma(\mathcal{H})/\mathcal{A}_\Gamma, \tag{56}$$

where $\mathcal{X}_{\mathrm{id}}$ contains independently established identification data.
The selector must satisfy:

S1. **Completeness.** A full grading or a consistent generative rule must be supplied.

S2. **Orthogonality.** The sectors must form a projective resolution of identity.

S3. **Affine-gauge respect.** The selector must not create a false absolute grade origin.

S4. **Relabel covariance.** Equivalent descriptions must produce gauge-equivalent gradings.

S5. **Operational identifiability.** The operational stabilizer must preserve the grading up to allowed relabeling.

S6. **Structurality.** The grading may not be independently refitted for each test state.

S7. **Domain compatibility.** Selected projectors must be compatible with the domain/form structure of H.

S8. **Projection compatibility.** Exact reduction must remain mathematically well defined.

S9. **No forced superselection.** The grading need not commute with H.

S10. **Data separation.** Identification data must be separated from target-test data.

S11. **Out-of-sample prediction.** At least one unused operational quantity must be predicted.

S12. **Non-generic relation.** The prediction must not follow from generic operator theory.

S13. **Comparator.** At least one serious alternative need not obey the relation.

S14. **Falsifiability.** Failure of the frozen relation must reject the proposed completion branch.

The mathematical envelope is nonempty. No physical mechanism within HSMT presently occupies it.

19 Why One Projector Is Not Enough

Suppose an experiment identifies one physical projector P. Then

$$Q = I - P \tag{57}$$

is fixed. However, $Q\mathcal{H}$ generally admits many inequivalent decompositions,

$$Q = \sum_j Q_j. \tag{58}$$

Thus one selected sector does not determine a complete grading, relative grade distances, adjacency, or an intergrade constitutive law. A physical completion therefore requires either an entire selected family or a rule that consistently extends to one.

20 Identification Data and Test Data

Future empirical construction must distinguish

$$\mathcal{D}_{\mathrm{id}} \tag{59}$$

from

$$\mathcal{D}_{\mathrm{test}}. \tag{60}$$

The grading is selected using

$$\mathcal{D}_{\mathrm{id}} \longrightarrow \mathscr{P}. \tag{61}$$

A non-generic HSMT relation must then predict an unused quantity:

$$\mathscr{P} \longrightarrow R_{\mathrm{HSMT}} \longrightarrow \widehat{\mathcal{O}}_{\mathrm{test}}. \tag{62}$$

One may not choose

$$\mathscr{P}_\star = \arg\min_{\mathscr{P}} \chi^2[R_{\mathrm{HSMT}}(\mathscr{P})] \tag{63}$$

from the same observations later presented as evidence for R_{HSMT}. That procedure is circular.

21 Post-Maturation Claim Hierarchy

21.1 Derived or frozen

The following survive under their declared assumptions:

- projectors follow from a declared grading;
- exact Feshbach reduction follows from H and P;
- bounded intergrade hopping admits self-adjointness control;
- projected dynamics contain exact self-energy/memory structure;
- absolute grade origin is not selected;
- grade and dimension are distinct;
- support incidence/exterior structure is canonical once support is declared;
- measured refinement requires explicit measure transport;
- the tested positive GC branch has a controlled vacuum runaway;
- P-CYCLE has spectrum (37);
- P-CYCLE is exactly scale neutral in the tested common-scale direction;
- analog cycle-mode multiplicity equals b_1;
- positive divergence-penalty changes preserve that mode;

- operational structure is identifiable only modulo the operational stabilizer;
- the MC-1 admissibility envelope is mathematically nonempty;
- MC-Core is generic affine-graded quantum mechanics.

21.2 Conditional

The following require additional declared assumptions:

- Gaussian intergrade coupling;
- nearest-neighbor dominance at the historical candidate width;
- dimensional characterization after a named observable is defined;
- primitive relational locality;
- metric–measure–energy realization;
- causal recovery after principal-symbol selection;
- matter-seed constructions;
- gravity-response realizations;
- measured relational refinement;
- P-CYCLE as an optional primitive;
- P-CYCLE analog implementation;
- any future natural bridge satisfying the frozen criteria.

21.3 Closed or nonselected

The following are not current derived HSMT conclusions:

- a multiplicity law;
- core selection of the historical 1/12 chain;
- target exponents derived from a fixed Gaussian width;
- IR Weyl selection of the special Gaussian width;
- a frozen physical selector for the historical $x = 16$ structure;
- a successful UV selector from the examined spectral-weighting routes;
- deterministic equivariant locality from fully symmetric support data;
- identification of grade with time;
- an intrinsic global arrow from finite recurrent closed unitary dynamics;
- a stable vacuum in the tested positive GC branch;

- rescue by q-independent positive refinement mixing;
- rescue by canonical positive Gaussian overlap;
- an independent reciprocal physical sector from the examined dualities;
- nonzero physical flux implied merely by cycle space;
- vacuum stabilization by scale-neutral P-CYCLE;
- unique physical content from pure measure reparametrization;
- a justified second new primitive after P-CYCLE;
- physical shell identification from ordinary contact spectroscopy alone;
- P-CYCLE excitation visibility through inherited ground-lift contacts;
- fundamental ontology discrimination from engineered linear analogs;
- a currently identified broad natural P-CYCLE bridge;
- a currently frozen core-level unique Nature-facing prediction;
- a complete HSMT grading from one selected projector;
- evidential value from a selector fitted to its own test relation.

22 Demoted Historical Claims

The following may appear only as historical motivation or conditional realizations:

- $N = 4$ as a physically selected absolute grade;
- $\rho_N = N - 4$ as a derived law;
- $\sigma_0 = \sqrt{2}/4$ as a uniquely derived physical constant;
- the historical $1/12$ chain as a core derivation;
- shell grade as physical dimension;
- the CMB as an established shell boundary;
- cosmological redshift as established dimensional progression;
- Type-Ia supernova dimming as established shell-energy leakage;
- dark matter or dark energy as identified higher-grade mechanics;
- Clifford structure as observed fermion content;
- internal algebraic structure as the Standard Model gauge group;
- multiplicity spaces as the observed three generations;

- overlap matrices as physical Yukawa matrices;
- scalar kinematics as the Higgs sector;
- Schur/seesaw structure as observed neutrino physics;
- the GC spectral branch as Einstein gravity.

23 What HSMT Currently Does and Does Not Explain

At its present maturity level HSMT supplies a controlled framework for graded sectors, microscopic cross-grade dynamics, exact projection into effective sectors, memory and self-energy generated by hidden sectors, relational support and locality modules, metric–measure–energy modules, controlled refinement constructions, conditional matter and gravity branches, explicit theorem provenance, and precise nonselection and no-go boundaries.

An important part of this content is negative: HSMT now identifies why several attractive mechanisms do not suffice.

HSMT does not presently derive

- the physical grading of Nature,
- absolute $N = 4$,
- the observed spacetime dimension,
- a unique Gaussian width,
- Standard Model gauge structure,
- three observed generations,
- physical Yukawa matrices,
- the Higgs sector,
- observed neutrino parameters,
- a stable geometric vacuum,
- Einstein gravity,
- dark matter,
- dark energy,
- cosmological redshift,
- the cosmic microwave background, or
- a unique Nature-facing HSMT prediction.

24 The Predictivity Gap

Predictivity gap. HSMT currently lacks a physically selected and operationally identifiable grading together with a non-generic measurable relation among its selected sectors.

This is stronger than saying merely that HSMT "needs experiments." Before a decisive experiment can exist, the framework must supply a quantity that is both HSMT-specific and operationally identifiable.

25 What Would Count as a Genuine Breakthrough?

A genuine future advance would do at least one of the following without target fitting:

1. supply an independently motivated MC-1-admissible grading selector;
2. derive a non-generic intergrade constitutive law from an independent physical principle;
3. construct an operational bridge that identifies the grading and leaves at least one out-of-sample prediction;
4. produce a stable vacuum mechanism from independently motivated new physics;
5. derive matter or gravity structure with enough specificity to exclude operational alternatives; or
6. identify an independently occurring natural system satisfying a frozen HSMT module law without having been engineered to emulate HSMT.

The following do not by themselves constitute breakthroughs: fitting existing data; reproducing a generic operator theorem; introducing a freely tunable function; renaming an ordinary physical variable as a shell quantity; restoring a selector already closed by nonselection; or engineering a device to implement equations deliberately programmed into that device.

26 Claim Hierarchy for Future Work

Future HSMT papers should separate claims into the following classes.

Class A: Generic substrate

Results derived only from MC-Core. No uniquely HSMT novelty claim should be attached to them.

Class B: HSMT architecture

Results requiring the combination

$$\text{graded geometry} + \text{intergrade coupling} + \text{projected effective sector}. \tag{64}$$

These define HSMT framework identity.

Class C: Module results

Results requiring a specific kernel, locality, metric, refinement, matter, or gravity assumption. The module assumptions must be stated explicitly.

Class D: No-go and nonselection results

Negative results are part of the mature theory and should be presented prominently.

Class E: Operational claims

Nature-facing statements must identify their preparations, controls, readout, selector, bridge, comparator, and falsification condition.

Class F: Optional extensions

New primitives such as P-CYCLE must remain clearly separated from HSMT-Core.

27 Claims Not Currently Licensed

Without a new derivation, future HSMT papers should not claim that

- HSMT derives four dimensions;
- HSMT derives $N = 4$;
- HSMT derives the special Gaussian width;
- HSMT derives the historical $1/12$ spectral chain;
- HSMT derives the Standard Model;
- HSMT derives Einstein gravity;
- HSMT explains dark matter;
- HSMT explains dark energy;
- HSMT explains cosmological redshift;
- HSMT explains the CMB;
- P-CYCLE solves the vacuum problem;
- the minimal HSMT core is a new form of quantum mechanics; or
- a successful engineered analog confirms fundamental HSMT.

These statements lie outside the frozen claim boundary.

28 Current Theory Status

Layer	Status	Interpretation
MC-Core	Derived	Generic affine-graded quantum-mechanical substrate.
$M_{\mathrm{GEO}} + M_{\mathrm{K}}$	Architectural	Minimum Level-1 HSMT framework.
Gaussian realization	Conditional	Level-2 constitutive model choice.
L0 / ME0	Primitive / conditional	Locality and metric–measure modules.
Matter seeds	Conditional	Mathematical possibilities, not Standard Model derivation.
GC gravity branch	Derived no-go	Controlled vacuum obstruction under declared assumptions.
P-CYCLE	Optional	Coherent structural primitive; scale neutral in the tested direction.
Operational grading selector	Open	Required for an empirically distinctive Level-3 completion.
Unique Nature-facing HSMT prediction	Open	Not presently established.

29 Discussion

The central outcome of the maturation programme is not a new numerical prediction. It is a reorganization of what the theory means.

Early HSMT development explored novelty simultaneously in hierarchy, dimension, Gaussian coupling, spectral scaling, time, matter, gravity, vacuum structure, and cosmology. The post-maturation theory separates these questions.

First, the generic mathematical substrate is acknowledged as generic. Second, HSMT architectural identity is located in grade-wise geometry, cross-grade dynamics, and projected effective sectors. Third, constitutive realizations are separated from the architecture. Fourth, optional primitives are separated from the core. Fifth, operational identification is treated as an independent scientific problem.

This organization prevents three recurrent errors:

1. mistaking generic mathematics for uniquely HSMT physics;
2. mistaking an architectural possibility for an empirical prediction;
3. mistaking a successful conditional branch for a completed fundamental theory.

The present framework is less ambitious in immediate empirical claims than earlier versions. Scientifically, that is progress. A theory whose unresolved assumptions are explicit can be improved and tested. A theory whose assumptions are hidden inside its conclusions cannot.

30 Conclusion

HSMT began as a broad attempt to explain familiar physics through a hierarchy of shell manifolds. The mature framework is considerably narrower.

Its theorem-minimal operator substrate is generic affine-graded quantum mechanics. Its nontrivial present identity lies in a modular architecture combining grade-wise geometric realization, cross-grade dynamics, and exact projection into effective sectors.

Several constitutive branches remain coherent. Several have produced useful no-go results. Several historically attractive identifications have been explicitly demoted. The optional P-CYCLE extension remains coherent but does not solve the tested vacuum problem and has no current natural physical bridge.

Most importantly, the empirical obstruction is now sharply localized: HSMT lacks a physically selected and operationally identifiable grading together with a non-generic measurable relation among the selected sectors.

That boundary determines the next genuine scientific task. A future breakthrough should not be judged by whether it adds complexity, reproduces a desired number, or fits an existing phenomenon. It should be judged by whether it closes this specific predictivity gap without reopening mechanisms already excluded by the maturation programme.

Current HSMT claim.
HSMT is a modular framework in which grade-wise geometric sectors are dynamically coupled and effective sector physics is obtained by exact projection; its generic operator substrate, module boundaries, and major no-go results are understood, while the physical grading and a unique operational intergrade law remain open.

Status Statement

At the completion of the post-maturation synthesis,

$$\text{HSMT} = \text{Level-1 modular framework} + \text{conditional Level-2 models}, \tag{65}$$

with no established Level-3 empirical completion.

The September public architecture remains unchanged. The generic substrate is understood. The architectural identity is explicit. Major failed selector searches remain closed. The predictivity gap is localized. The future admission standard is defined. No stronger physical claim is made here.

A Claim-Status and Provenance Notation

To prevent future ambiguity, the following notation is recommended.

Label	Meaning
`[D]`	Derived under explicitly stated assumptions.
`[C]`	Conditional on an additional model or module assumption.
`[P]`	Primitive assumption or declared structural input.
`[NG]`	No-go, nonselection, or negative closure result.
`[DEM]`	Historical claim demoted from derived status.
`[EXT]`	Optional extension not contained in HSMT-Core.
`[OPEN]`	Unresolved question or missing physical bridge.

For theorem provenance, use:

Tag	Provenance
[MC]	Theorem-minimal generic graded-quantum substrate.
[GEO]	Grade-wise geometric realization.
[K]	Restricted intergrade coupling law.
[L]	Relational locality module.
[ME]	Metric–measure–energy module.
[R]	Refinement module.
[GR]	Gravity-response module.
[MAT]	Matter/internal-algebra module.
[OP]	Operational bridge.
[EXT]	Optional primitive extension such as P-CYCLE.

A theorem using only [MC] must not be described as uniquely HSMT. A theorem using one or more additional modules must state those assumptions explicitly.

B Compact Post-Maturation Claim Ledger

Status	Topic	Current claim
[D]	MC-Core	Equivalent to generic affine-graded quantum mechanics.
[D/C]	HSMT architecture	Grade-wise geometry plus intergrade dynamics plus exact projected effective sectors define the current framework identity.
[C]	Gaussian kernel	A valid constitutive realization, not selected by MC-Core.
[DEM]	$\sigma_0 = \sqrt{2}/4$	Historical candidate with nearest-neighbor dominance in its declared realization; not a derived physical constant.
[DEM]	$N = 4$	Not a physically selected absolute grade.
[DEM]	Grade = dimension	Explicitly rejected.
[NG]	Locality selection	Fully symmetric support does not deterministically select a preferred locality relation.
[NG]	Intrinsic arrow	Finite recurrent closed unitary dynamics does not generate a global monotone arrow.
[NG]	GC vacuum	The tested positive spectral branch has no finite unconstrained stationary point.
[EXT]	P-CYCLE	Coherent optional primitive; metric-independent spectrum and exact scale neutrality in the tested common-scaling direction.
[NG]	P-CYCLE vacuum role	Closed: scale neutrality prevents stabilization of the tested GC runaway.
[NG]	Analog ontology	A sufficiently general ordinary oscillator realization can reproduce the same linear-response data.

Status	Topic	Current claim
[OPEN]	Physical grading	No operationally selected grading currently exists.
[OPEN]	Nature-facing prediction	No uniquely HSMT, operationally identified Level-3 prediction is presently established.

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